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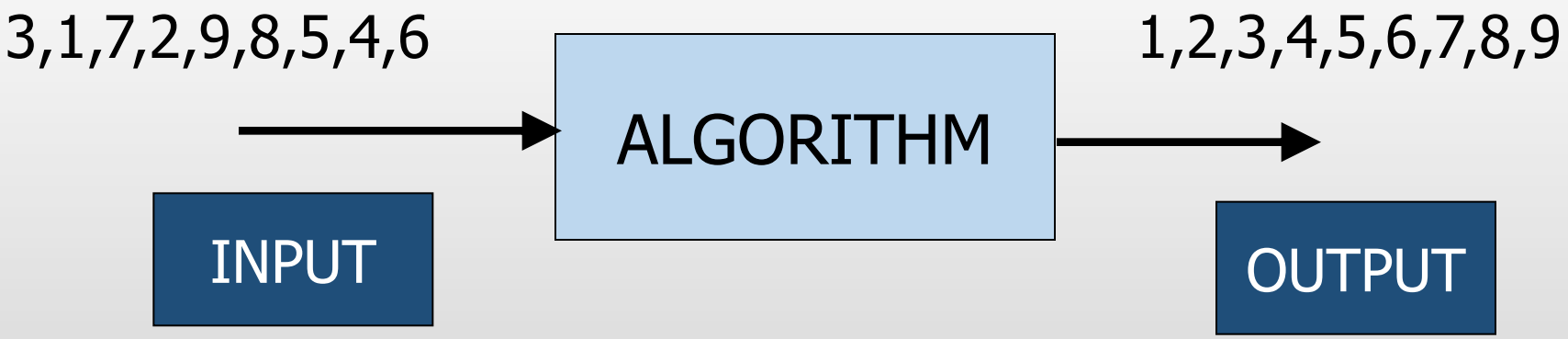
# IT1170 – Data Structures and Algorithms

## Lecture 01

## Introduction to Algorithms

# ALGORITHMS

- Algorithm is any well defined computational procedure that takes some value or set of values as input and produce some value or set of values as output.

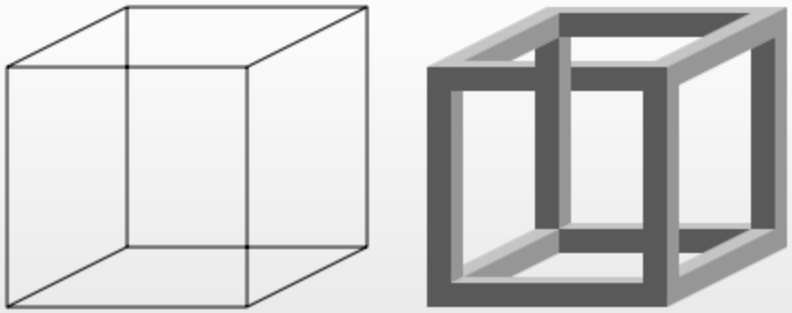


# ALGORITHM (Contd.)

1. Get the smallest value from the input.
2. Remove it and output.
3. Repeat above 1,2 for remaining input until there is no item in the input.

# Properties of an Algorithm.

- Be correct.
- Be unambiguous.
- Give the correct solution for all cases.
- Be simple.
- It must terminate.



Necker_cube_and_impossible_cube

Source:http://en.wikipedia.org/wiki/Ambiguity#Mathematical_interpretation_of_ambiguity

# Applications of Algorithms

- Data retrieval
- Network routing
- Sorting
- Searching
- Shortest paths in a graph

# Pseudocode

- Method of writing down a algorithm.
- Easy to read and understand.
- Just like other programming language.

- More expressive method.
- Does not concern with the technique of software engineering.

# Pseudocode Conventions.

- English.
- Indentation.
- Separate line for each instruction.
- Looping constructs and conditional constructs.
- // indicate a comment line.
- = indicate the assignment.

# Pseudocode Conventions.

- Array elements are accessed by specifying the array name followed by the index in the square bracket.
- The notation “..” is used to indicate a range of values within the array.

Ex:

A[1..i] indicates the sub array of A consisting of elements A[1] , A[2] , .. , A[i].

# Analysis of Algorithms

Idea is to predict the resource usage.

- Memory
- Logic Gates
- **Computational Time**

Why do we need an analysis?

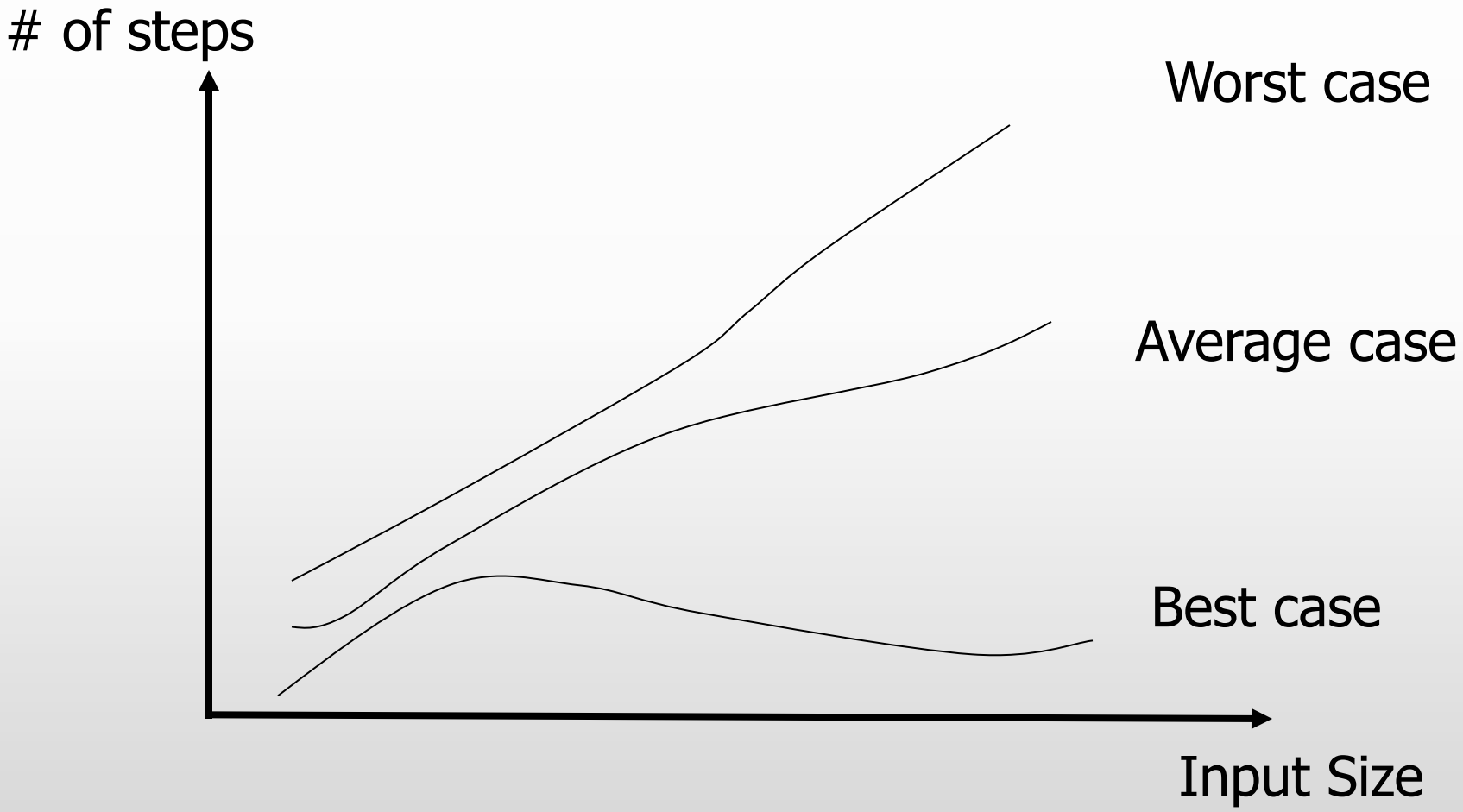
- To compare
- Predict the growth of run time

# Worst, Best and Average case.

Running time will depend on the chosen instance characteristics.

- **Best case**:
  Minimum number of steps taken on any instance of size n.
- **Worst case:**
  Maximum number of steps taken on any instance of size n.
- **Average case:**
  An average number of steps taken on any instance of size n.

# Worst,Best and Average case(Contd.)



# Analysis Methods

- Operation Count Methods
- Step Count Method(RAM Model)
- Exact Analysis
- Asymptotic Notations

# Operation count

- Methods for time complexity analysis.
- Select one or more operations such as add, multiply and compare.
- Operation count considers the **time spent on chosen operations** but not all.

# Step Count (RAM Model)

- Assume a generic one processor.
- Instructions are executed one after another, with no concurrent operations.
- +, - , =, it takes exactly one step.
- Each memory access takes exactly 1 step.
- **Running Time = Sum of the steps.**

# RAM Model Analysis.

Example1:

| | |
|---|---|
| n = 100 | 1step |
| n = n + 100 | 2steps |
| Print n | 1step |

Steps = 4

Example2:

$sum = 0$ — 1 assignment

$\text{for } i = 1 \text{ to } n$ — $n$+1 assignments, $n$+1 comparisons, $n$ additions

$sum = sum + A[i]$ — n assignments, n additions, n memory accesses

Steps = 6n+3

# Question 01

- Using RAM model analysis, find out the no of steps needed to display the numbers from 1 to 10.

i = 1 → 1 step

While i <=10 → 11 steps

print i → 10 steps

i = i + 1 → 10 + 10 = 20 steps

Steps = 42

# Question 02

- Using RAM model analysis, find out the no of steps needed to display the numbers from 10 to 20.

i = 10 → 1 step

While i <= 20 → 12 steps ( Hint :20 – 10 + 2 = 12)

print i → 11 steps

i = i + 1 → 11 + 11 = 22 steps

Steps = 46

# Question 03

- Using RAM model analysis, find out the no of steps needed to display the even numbers from 10 to 20.

for i = 10 to 20 → (12+ 12 + 11) steps = 35 steps

if i % 2 == 0 → 2 * 11 = 22 steps

print i → 6 steps

Steps = 63

# Problems with RAM Model

- Differ number of steps with different architecture.

  eg: sum = sum + A[i]  is a one step in  the CISC processor.

- It is difficult to count the exact number of steps in the algorithm.

  eg: See the insertion sort , efficient algorithm for sorting small number of elements.

# Summary

- What is an algorithm?
- Properties of an algorithm.
- Design methods.
- Pseudocode.
- Analysis(Operation count & Step count, RAM model).